

VEDA Technology Internship — Task 6 Report

BMI Calculator Application Documentation

1. Project Overview

This report details **Task 6 (BMI Calculator)** for the VEDA Technology Web Development Internship. The web application allows users to enter height (in cm) and weight (in kg), calculate their Body Mass Index (BMI), and categorize their result into Underweight, Normal Weight, Overweight, or Obese.

2. Technical Stack

Component	Technology Details
HTML5	Semantic structure (<header>, <main>, <section>, <form>, <label>, <input>, <button>, <footer>)
CSS3	Custom Soft Mint & White theme (--bg: #F3FAF7, --primary: #5FAF92, --light-accent: #DDF1E8, --border: #D5E8DF)
Vanilla JS	DOM selection, form submission, input validation rules, BMI formula, if/else categorization

3. BMI Formula & Categories

Formula	Range / Boundary	Category Output
BMI = weight (kg) / height (m) ²	Below 18.5	Underweight
Convert height from cm to meters	18.5 – 24.9	Normal Weight
Round output to 1 decimal place	25.0 – 29.9	Overweight
E.g. 70 kg & 175 cm → 22.9	30.0 or above	Obese

4. Input Validation Rules

- **Empty Input Validation:** Displays *"Please enter your height and weight."*
- **Non-Numeric Validation:** Displays *"Please enter valid numeric values."*
- **Height Boundary (≤ 0):** Displays *"Height must be greater than 0."*
- **Weight Boundary (≤ 0):** Displays *"Weight must be greater than 0."*

5. Verification & Compliance Checklist

- Verified normal BMI (e.g., 70kg, 175cm → 22.9, Normal Weight)
- Verified underweight (e.g., 50kg, 175cm → 16.3, Underweight)
- Verified overweight (e.g., 85kg, 175cm → 27.8, Overweight)
- Verified obese (e.g., 100kg, 175cm → 32.7, Obese)
- Verified empty, zero, negative, and decimal inputs & Reset functionality